Tutorial - 5

① (a)

$$\psi(r,0) = \frac{\sqrt{2}}{2\sqrt{L^3}}\left[e^{i2\pi x/L} + e^{-i2\pi x/L}\right] e^{i(3\pi y + 5\pi z)/L}$$

$$= \frac{1}{\sqrt{2L^3}}\left[e^{i(2\pi x + 3\pi y + 5\pi z)/L} + e^{i(3\pi y + 5\pi z - 2\pi x)/L}\right]$$

$$= \frac{1}{\sqrt{2}}\left[\underbrace{\frac{1}{\sqrt{L^3}} e^{i\bar{k}_1\cdot\bar{r}}}_{|p_1\rangle} + \underbrace{\frac{1}{\sqrt{L^3}} e^{i\bar{k}_2\cdot\bar{r}}}_{|p_2\rangle}\right].$$

$\bar{k}_1 = \frac{\pi}{L}(2\hat{x} + 3\hat{y} + 5\hat{z})$; $\bar{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$\bar{k}_2 = \frac{\pi}{L}(-2\hat{x} + 3\hat{y} + 5\hat{z})$

$\psi(r,0)$ is a superposition of 2 momentum eigenstates

$= \frac{1}{\sqrt{2}}|p_1\rangle$ & $|p_2\rangle$.

Why $|p_1\rangle$ & $|p_2\rangle$ are eigenstates of $\hat{p}$?

~~Pr Any~~ $\hat{p} = -i\hbar\nabla$; $\bar{k} = k_x\hat{x} + k_y\hat{y} + k_z\hat{z}$

$\bar{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$\therefore e^{i\bar{k}\cdot\bar{r}} = e^{i(k_x x + k_y y + k_z z)} = \psi(r)$.

$$\therefore \hat{p}\,\psi(r) = -i\hbar\left(\frac{\partial}{\partial x}\hat{x} + \frac{\partial}{\partial y}\hat{y} + \frac{\partial}{\partial z}\hat{z}\right)\psi(r)$$

$$= -i\hbar\left[k_x\hat{x} + k_y\hat{y} + k_z\hat{z}\right] e^{i\bar{k}\cdot\bar{r}}$$

$$\boxed{\hat{p}\,\psi(r) = \hbar\bar{k}\,\psi(r)}$$

Thus, $\psi(r) = e^{i\bar{k}\cdot\bar{r}}$ is an eigenstate of $\hat{p}$.

Let $\psi_n(x) = A e^{i\bar{k}\cdot\bar{r}}$ be the normalized wavef^n.

$$\therefore \int_0^{L^3} |\psi_n(x)|^2\, dx = 1, \quad (3D)$$

$$\Rightarrow A^2 = \frac{1}{L^3} \Rightarrow \boxed{A = \frac{1}{\sqrt{L^3}}}$$

$$\therefore \boxed{\psi_n(x) = \frac{1}{\sqrt{L^3}} e^{i\bar{k}\cdot\bar{r}}}$$

Possible measurement outcomes :

$|p_1\rangle$: $p_1 = \hbar k_1 = \frac{\pi\hbar}{L}(2\hat{x} + 3\hat{y} + 5\hat{z})$. [Ans]

$|p_2\rangle$: $p_2 = \hbar k_2 = \frac{\pi\hbar}{L}(-2\hat{x} + 3\hat{y} + 5\hat{z})$.

~~Prob~~ $\therefore \psi(r,0) = \frac{1}{\sqrt{2}}[|p_1\rangle + |p_2\rangle]$.

Probability of measurement is the square of the absolute value of the coefficients of the states.

$|p_1\rangle$: $P(p_1) = |\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$ [Ans]

$|p_2\rangle$: $P(p_2) = |\frac{1}{\sqrt{2}}|^2 = \frac{1}{2}$

(b) For a free particle :

$$\hat{H} = \frac{\hat{p}^2}{2m}$$

$\therefore [\hat{H}, \hat{p}] = 0$.

The momentum eigenstates are also the eigenstates of $\hat{H}$.

$\therefore E = \frac{\hbar^2|k|^2}{2m}$ — energy for a free particle

$\therefore E_1 = \frac{\hbar^2|k_1|^2}{2m} = \frac{\hbar^2}{2m} \cdot \frac{38\pi^2}{L^2} = \frac{19\pi^2\hbar^2}{mL^2}$

$\therefore E_2 = \frac{\hbar^2|k_2|^2}{2m} = \frac{\hbar^2}{2m} \cdot \frac{38\pi^2}{L^2} = \frac{19\pi^2\hbar^2}{mL^2}$] Degenerate Energies.

$\therefore \psi(r,t) = \psi(r,0) \cdot e^{-iEt/\hbar}$

$= \frac{\sqrt{2}}{\sqrt{L^3}} \cos\left(\frac{2\pi x}{L}\right) e^{i(3\pi y + 5\pi z)/L} \cdot e^{-i\frac{19\pi^2\hbar^2}{mL^2}t}$ (Ans)

(c) $\vec{p}$ is the eigenvalue of $|p_1\rangle$.

$\therefore \psi(r,0^+) = \frac{1}{\sqrt{L^3}} \cdot e^{i\vec{k_1}\cdot\vec{r}}$

[The wave fn collapses to the state $|p_1\rangle$].

Time Evolutn :

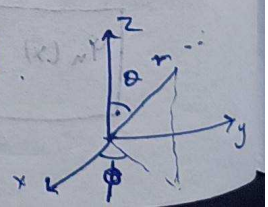
$\psi(r,t) = \psi(r,0^+) \cdot e^{-iEt/\hbar}$

$= \frac{1}{\sqrt{L^3}} \cdot e^{+i(2\pi x + 3\pi y + 5\pi z)/L} \cdot e^{-i\frac{19\pi^2\hbar^2}{mL^2}t}$ (Ans)

② (a) Inner Product is calculated by integrating Y_{lm} with $d\Omega$

$d\Omega = \sin\theta \, d\theta \, d\phi$.

$[\, d\Omega = \frac{dA}{r^2} = \frac{(r\,d\theta)(r\sin\theta\,d\phi)}{r^2} = \sin\theta \, d\theta \, d\phi \,]$



$$\therefore \langle Y_{00} | Y_{00} \rangle = \int \frac{1}{4\pi} \int_0^{\pi} \sin\theta \, d\theta \int_0^{2\pi} d\phi .$$

$$= -\frac{1}{4\pi} \cdot 2\pi \cdot \cos\theta \Big|_0^{\pi} = \frac{1}{2} \cdot 2 = 1 .$$

$$\therefore \langle Y_{10} | Y_{10} \rangle = \frac{3}{4\pi} \int_0^{\pi} \cos^2\theta \sin\theta \, d\theta \int_0^{2\pi} d\phi$$

$$= \frac{3}{4} \int_0^{\pi} \sin 2\theta \, d\theta \cos\theta$$

$$\cancel{= \frac{3}{4} \left[\frac{\cos 2\theta}{2} \right]_0^{\pi} = \frac{3}{8}} .$$

$$= \frac{3}{8} \int_0^{\pi} \sin 3\theta + \cancel{\cos\theta} \sin\theta \, d\theta .$$

$$= \frac{3}{8} \left[-\frac{\cos 3\theta}{3} + \cancel{\sin\theta} \cos\theta \right]_0^{\pi}$$

$$= \frac{3}{8} \left[2 + \frac{1}{3} + \frac{1}{3} \right] = 1 .$$

Alternative :

$$\int_0^{\pi} \cos^2\theta \sin\theta \, d\theta$$

Let $\cos\theta = t \Rightarrow -\sin\theta \, d\theta = dt$.

$$\therefore \int_0^{\pi} \cos^2\theta \sin\theta \, d\theta = \int_{-1}^{1} t^2 \, dt = \frac{2}{3} t^3 \Big|_0^1 = \frac{2}{3}$$

$$\therefore \langle Y_{10} | Y_{10} \rangle = \frac{3}{4\pi} \cdot \int_0^{\pi} \cos^2\theta \sin\theta \, d\theta \int_0^{2\pi} d\phi$$

$$= \frac{3}{4\pi} \cdot \frac{2}{3} \cdot 2\pi = 1$$

$$\therefore \langle Y_{10} | Y_{00} \rangle = \frac{\sqrt{3}}{4\pi} \int_0^{\pi} \cos\theta \sin\theta \, d\theta \int_0^{2\pi} d\phi$$

$$= \frac{\sqrt{3}}{4} \int_0^{\pi} \sin 2\theta \, d\theta = -\frac{\sqrt{3}}{4} \left[\frac{\cos 2\theta}{2} \right]_0^{\pi}$$

$$= 0$$

Thus, they are orthonormal

(b) Y_{00} :

$$\therefore \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2} Y_{00} = 0 .$$

$$\therefore \hat{L}^2 Y_{00} = 0$$

$$\therefore \sin\theta \frac{\partial}{\partial\theta} Y_{00} = 0 .$$

$$\therefore \hat{L}_z Y_{00} = 0 .$$

Eigenvalue $= 0$. (Ans)

Y_{10} :

$$\therefore \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2} Y_{10} = 0$$

$$\frac{1}{\sin\theta} \frac{\partial}{\partial\theta} \left(\sin\theta \frac{\partial}{\partial\theta} Y_{10} \right) = -\sqrt{\frac{3}{4\pi}} \cdot \frac{1}{\sin\theta} \frac{\partial}{\partial\theta} (\sin^2\theta)$$

$$= -\sqrt{\frac{3}{4\pi}} \cdot \frac{1}{\sin\theta} \sin\theta \cos\theta = -\sqrt{\frac{3}{4\pi}} \cos\theta$$

$\therefore \hat{L}^2 Y_{10} = 2\hbar^2 Y_{10}$. —— Eigenvalue : $2\hbar^2$

$\therefore \hat{L}_z Y_{10} = -i\hbar \frac{\partial}{\partial \phi} Y_{10} = 0$. —— Eigenvalue : 0

(3) $\hat{H} |l, m\rangle = E |l, m\rangle$.

$\Rightarrow \left(\frac{L^2}{2I} + \alpha L_z\right) |l, m\rangle = \left(\frac{l(l+1)\hbar^2}{2I} + \alpha m \hbar\right) |l, m\rangle$.

$\therefore \boxed{E = \frac{l(l+1)\hbar^2}{2I} + \alpha m \hbar}$ (Ans) ; $l = 0, 1, 2, \ldots$

$m = -l, -l+1, \ldots 0, 1, \ldots l$